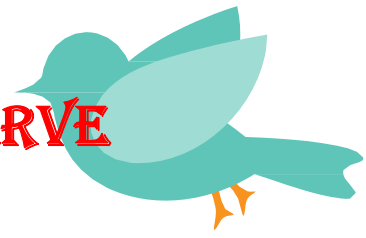


EXAMINATION OF PERIPHERAL NERVE INJURIES



NAME :-

AGE :-

SEX :-

OCCUPATION :- Agriculture (Leprosy), shipyard (lead poisoning)

CHIEF COMPLAINTS :-

1. Inability to perform certain movements:- eg holding a glass of water(claw hand),unable to dorsiflex his ankle, or walk properly(foot drop),unable to lift his wrist and fingers (wrist drop , finger drop).
2. weakness :- In leg, wrist, fingers etc
3. Deformity :- claw hand, wrist drop, foot drop,

HISTORY OF PRESENTING ILLNESS:- Elaborate on his complaints like

PAIN- ; (remember **SOCRATES** –SITE ,ONSET, CHARECTER, RADIATION , ASSOCIATION,TIMING ,EXCERBATING AND RELEVING FACTORS ,SEVERITY)

DEFORMITY- SITE, ONSET, DURATION, PROGRESSIVE OR NON PROGRESSIVE, ANY CORRECTION ATTEMPTED

when did he notice it, was it sudden in onset(compression neuropathy, Saturday night palsy,),or gradual(leprosy),was it unilateral (most of nerve injuries,) B/L(GB syndrome),Duration

ADL ; - (Activities of daily living);-holding glass of water, eating food on his own, difficulty in cycling, squatting,

NEGATIVE HISTORY:- Anesthetic patches over the back (leprosy),Trauma(fibular neck #,tibial condyle #s, compartment syndrome),Injections(palsies),Tumor(exostoses), following Surgery(acetabular#, IVDP) Low back pain(IVDP),Exposure to paints, working in ship yard(lead poisoning-motor neuropathy), Diabetes mellitus (mononeuropathy),Alcoholism, Using crutches(Saturday night palsy),sitting cross legged, vitamin deficiency – beriberi, Heavy metal poisoning(arsenic, , antimony-mixed neuropathy lower extr more involved),Drugs(INH, streptomycin, ethambutol, vincristine, cisplatin),OP poisoning,

MND (but have fasciculation's with muscle weakness, GB Syndrome(Ascending weakness, bilateral).

LOCAL EXAMINATION:-



THE EXAMINATION IS SLIGHTLY DIFFERENT IN NERVE INJURIES, DO NOT FORGET



{ ABCDEFGHIJK }---A for INSPECTION (attitude and deformity), B for palpation, C for muscle power, D for sensation, E for reflexes, F for vasomotor, G for movements H for measurements, I for muscles available for future tendon transfers, J for special tests for the particular nerve, k for vascular and lymph node examination.}

GAIT:- IS it Bipedal, aided or unaided, stable or unstable (high stepping), Coordinated, is it painful (antalgic)

ATTITUDE AND DEFORMITY :- patient in sitting posture, or supine posture, and describe the deformity like it's a wrist drop, or claw hand deformity, foot drop deformity

Pearls - what determines deformity- In paralytic conditions- the overpowering muscles determines the deformity,

In non paralytic conditions- Innate tendency of postural fixity in possible position of walking that determines the deformity.

A) INSPECTION:- if its **upper limb** nerve involvement, start with level of both shoulders, elbow position, wrist and fingers position, (eg; in a case of wrist drop, the shoulders are at same level and in neutral, elbow in 30 deg flexion, wrist in 90 deg palmar flexion, fingers in 45 deg flexion.) (pointing index-median nerve involvement) (flexion adduction and internal rotation of shoulder with forearm in pronation, fingers in flexion-erbs palsy)

if in **lower limb** nerve involvement start with (eg;- in foot drop case, ASIS are at same level, hip in extension, knee in extn, foot in 15 deg plantar flexion)

findings:-

1. Look for wasting of muscles (eg; 1st metacarpal space wasting in ulnar nerve involvement, and wasting of forearm extensor muscles in radial nerve involvement, wasting of calf muscles in foot drop,

2. Look for skin; make sure to comment on Dryness of skin, Brittle nails, loss of hairs, any scars, sinuses.

{B}:-PALPATION:-

1. Any Local rise in temperature (usually cold compared to opp limb)

2. Tenderness-superficial and deep

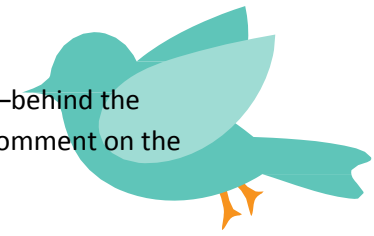
3. Muscle Bulk (say its less compared to normal side if present)

4. Tone (suppleness)

5. palpation of Nerve proper ; - **{ T T T B I }** ie {Tenderness, Thickening, Tinel's sign, Beading, Irritability} don't forget



(palpate along the course of nerve where its supfl and palpable (eg; for ulnar nerve –behind the medial epicondyle in elbow, for common peroneal-around the neck of fibula, and comment on the findings below)



Look for T T T B I ie :-

Thickening :- of the nerve seen in leprosy

Tenderness :- if nerve is tender its neuritis, seen in leprosy

Tinels sign- is it present or absent , (you should know about full tinel sign as long note , like definition, how to elicit, below to above, is it a high or low Tinels sign, progressive or static Tinels sign, and lot more(*JBJS-VOL 60-A,NO 3 APRIL 1978-ARTICLE ON TINELS SIGN* : PLS, READ ,BYHEART,EAT, DRINK,SWALLOW TINELS SIGN OK)

Beading ; -also seen in leprosy cause of nerve abscess

Irritability :- seen in neuritis, it twitches with shooting pain, also seen in cubitus valgus deformity.

{ C } :-MUSCLE POWER:- MAKE A MUSCLE CHART OF THE MUSCLES SUPPLIED BY THAT NERVE WITH ITS GRADE. Pls feel for the contraction of that particular muscle which you are testing to prevent TRICK movement(ie-one muscle will contract and do the job of other muscle –visually you will be fooled if u don't feel for the contraction)and to know the difference between power grade 1 and 2, u have to feel the contraction.

(eg; Radial Nerve –supplies triceps, anconeus,brachialis,brachioradialis,supinator, extensor group of fore arm muscles(ECRL and ECRB) and finger extensor s(Apl, Epb, extn digitorum,)sensation in 1st webspace.

Eg; Median Nerve;-In FA-pronator teres, FCR, PL, FDS, FDP, FPL,PQ.(all flexors except FCU),In Hand- APB,FPB, Middle and index lumbricals, sensation to radial 3 1/2 fingers.

Eg; Ulnar Nerve-in arm no supply, at elbow sensn to joint,Fcu,FDP(med ½), hypothenar, dorsal and palmar interossei, medial 2 lumbricals, and adductor polices. sensn to volar aspect of little finger, medial half of ring finger.

Eg; Common peroneal nerve;-divides in to supfl and deep peroneal N, supfl peroneal runs and supplies peroneal muscles and sensn in lateral border of foot, the deep supplies antr compartment(4 muscles) and supplies 1st web space.

Muscle Power {Grading is ; -0 – no contraction, 1 – flickr of contraction,2-contraction with gravity eliminated,3-against gravity,4-with slight resistance,5-full resistance}

{ D }:-SENSATION.

1. Tactile sensation-look for pain(use pin),light touch(use cotton wool),pressure,2 point discrimination(use compass)





2. Deep pain

3. Temperature;- use test tubes (hot and cold)

4. Vibration-use 256 HZ(they will ask u about which hertz you used –say 256 hz even if u don't have it)

5. Proprioception;-position of the joint,(ask him to close eyes and tell whether joint is flexed or extended.

6 . Stereognosis;-Recognition of size and shape of an object (use a keychain,pen, ask him to close eyes and identify the object.

{E};-REFLEXES ;- Superficial and Deep.(in upper limb- biceps, supinator,)(lower limb- knee jerk, ankle jerk). Pls learn about how to elicit the reflex and its grading (

{F};-VASOMOTOR;- Starch –iodine test(sweat test)

Guttman test –quinnarazine powder,

{G};-MOVEMENTS;-

ACTIVE

AND PASSIVE- KNOW ALL ROM OF ALL JOINTS , do it on a joint above and a joint below .

{ H };-MEASUREMENTS;-

LINEAR: – (length of particular limb, total and segmental measurement)

CIRCUMFERENTIAL :- (To know the wasting of muscles)

DEFORMITY ASSESEMENT ONLY IN CLAW HAND – Look for

Assisted angle –read in green text book of orthopedics vol -2(home work)

Contracture angle - read in green text book of orthopedics vol -2(home work)

{ I } make a note of available muscles for future tendon transfer plans .

{ I };- SPECIAL TESTS;- FOR THAT PARTICULAR NERVE

RADIAL NERVE ;--- screening test –inability to extend the thumb hitch hikers sign.

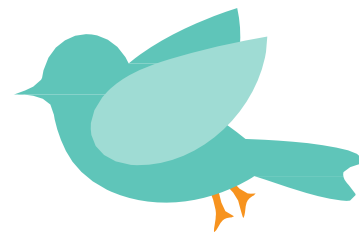
{First look for brachioradialis- to know high or low radial nerve palsy(if BR is intact its low nerve palsy ,if BR is lost its high radial nerve palsy]

1. brachioradialis- extension of forearm in mid pronation against resistance.

2.extensors of wrist-ECRL and ECRB-ask to extend wrist against resistance

3.extensor digitorum- extend MTP joint against resistance.





ULNAR NERVE ;-screening test – Loss of pain at tip of little finger.

{ If FCU & FDP medial ½ is involved it means it's a high ulnar nerve palsy }

Ulnar paradox;- higher the lesion less is the deformity that's obvious.

Froments sign (Book test);-1st palmar interossei, add polices, and FpL are required to hold a book, in ulnar n injury the 1st two muscles are paralysed now pt relies on fpl to hold book which appears prominent hence the test is positive.

Card test ;- loss of adduction due to paralysis of palmar interossei supplied by ulnar nerve.

Egawa test;-place palm on table move middle finger side wards, if not possible cos loss of dorsal interossei.

MEDIAN NERVE; - screening test- loss of sensation at tip of index finger.

PEN TEST; - Unable to touch the pen due to loss of abductor polices brevis,

POINTING INDEX (OSCHNER CLASP TEST) Long flexors of middle and index supplied by median nerve fail to flex when both hands clasped to gather.

DIAGNOSIS; ▲ ANATOMICAL –WHICH NERVE IS INVOLVED

PATHOLOGICAL-Neuropraxia, axonotemesis, neurotemesis, neuropathy, neuritis

INVESTIGATIONS ;-

LABORATORY;- Hb, Wbc, TcDc, ESR, CRP

RADIOLOGICAL ;- x ray of the part to rule out fractures

SPECIAL INVESTIGATIONS;-

Nerve conduction velocities(read and by heart)

Electromyography,(read and by heart)(normal activity, positive waves, denervation potentials ,SD curve)

Tests for leprosy- nasal scrapings, ear lobe biopsy, Blood levels for lead .



SELF NOTES

